

Stein Coefficients and their application to KMT Embedding

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Stein Coefficient

Stein identity for normal: $Z \sim \mathcal{N}(0, \sigma^2)$ iff for all 'sufficiently smooth' function f ,

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Modify for normal approximations:

$$E[Wf(W)] = E[Tf'(W)]$$

for W with mean zero and variance σ^2 , not necessarily normal.

We call T a 'Stein coefficient' for W . See Cacoullos and Papathanasiou 1992.

The idea is when T is close to σ^2 , W is close to $\mathcal{N}(0, \sigma^2)$.

Compare to zero bias transformation (Goldstein and Reinert 1997)

$$E[Wf(W)] = E[\sigma^2 f'(W^*)],$$

we see

$$E[T|W] = \sigma^2 \frac{d\mu^*}{d\mu}.$$

Since W^* has W -zero bias distribution, it always has a density. Thus, T cannot exist when the distribution of W has no absolutely continuous component.

Generally Stein coefficients are not so easy to construct.

References:

Bhattacharjee, C; Goldstein, L. On strong embeddings by Stein's method. Electron. J. Probab. 21 (2016).

Chatterjee, S. A new approach to strong embeddings. Prob. Theory and Related Fields. 152 (2012).

Coupling a Random Walk to Brownian Motion

Let $\epsilon_1, \epsilon_2 \dots$ be i.i.d. and distributed as ϵ , a mean zero, variance one random variable and for $1 \leq k \leq n$ and $t \in [0, 1]$, let

$$S_k = \sum_{i=1}^k \epsilon_i \quad \text{and} \quad X_n(t) = \frac{1}{\sqrt{n}}(S_{[nt]} + (nt - [nt])\epsilon_{[nt]+1}).$$

Then $X_n(t)$ converges weakly to a Brownian Motion $(B_t)_{t \geq 0}$ on $[0, 1]$ by Donsker.

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Goal is to find the best (smallest) rate of growth for $f(n)$ and a coupling between the processes so that

$$\max_{0 \leq k \leq n} |S_k - B_k| = O_p(f(n)).$$

Early Developments

Skorokhod 1961, Strassen 1966 used Skorokhod embedding to achieve the rate $(n \log \log n)^{1/4} (\log n)^{1/2}$ and conjectured this is the best possible rate when $E\epsilon^4 < \infty$.

Kiefer 1969 proved the conjecture.

Bártfai 1966 proved $\log n$ is best possible rate under finite moment generating function assumption.

Csörgő and Révész 1974/75 showed quantile transform method yields better rates under higher moment assumption.

Developing their ideas further, Komlós, Major and Tusnády (KMT) 1975 achieved the $\log n$ rate.

Strong Embedding (KMT)

We say strong embedding (SE) holds for $\mathcal{L}(\epsilon)$ if there exist positive constants C, K and λ such that for all n

$$P\left(\max_{0 \leq k \leq n} |S_k - B_k| \geq C \log n + x\right) \leq Ke^{-\lambda x} \quad \text{for all } x \geq 0.$$

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Komlós, Major and Tusnády (KMT) 1975 showed SE holds for $\mathcal{L}(\epsilon)$ with finite m.g.f. in a neighborhood of zero.

Numerous applications, see the texts of Shorack and Wellner, 1986, or Csörgő and Révész, 1981.

Uses coupling argument which are quite complicated and hard to follow.

Ultimate goal is to extend to dependent variables.

Using Stein's Method

Chatterjee 2012 introduced a novel approach using ideas from Stein's Method to show SE holds for Rademacher summands.

And 'collateral' new result that for $\epsilon_1, \dots, \epsilon_n$ exchangeable variables taking values in $\{-1, +1\}$, there exist a coupling of

$$W_k = S_k - \frac{k}{n} S_n$$

and $(B_t)_{t \in [0,1]}$, a standard Brownian Bridge, and positive constants λ_0, C and K such that for all $\lambda \leq \lambda_0$

$$E \exp(\lambda \max_{0 \leq k \leq n} |W_k - \sqrt{n} B_{k/n}|) \leq E \exp \left(C \log n + \frac{K \lambda^2 S_n^2}{n} \right).$$

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T encodes all information regarding closeness of W to $\mathcal{N}(0, \sigma^2)$.

Theorem 1.2, Chatterjee 2012

Let W be a variable mean zero with finite second moment and suppose that T is a Stein coefficient for W with $|T|$ almost surely bounded by a constant. Then, given any $\sigma^2 > 0$, we can construct a version of W and $Z \sim \mathcal{N}(0, \sigma^2)$ on the same probability space such that

$$E \exp(\theta |W - Z|) \leq 2E \exp\left(\frac{2\theta^2(T - \sigma^2)^2}{\sigma^2}\right) \quad \text{for all } \theta \in \mathbb{R}.$$

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Note that bound is tight in that $T = \sigma^2$ implies $W \sim \mathcal{N}(0, \sigma^2)$, and that $\sigma^2 > 0$ is arbitrary (not necessarily equal to variance of W).

Sampling Random Walk

Sample the random walk in a convoluted manner:

- 1 Sample S at the terminal time n of the random walk.
- 2 Sample a multiset $\{\epsilon_1, \dots, \epsilon_n\}$ of independent $\mathcal{L}(\epsilon)$ variables conditional on their sum being S .
- 3 Now vary about the line connecting $(0,0)$ to (S,n) by sampling an independent, uniformly random permutation π , and forming the random walk

$$S_k = \sum_{j=1}^k \epsilon_{\pi(j)}.$$

Coupling Overview

- 1 Sample the joint values S and $Z \sim \mathcal{N}(0, n)$ of the coupled random walk and Gaussian path at time n .
- 2 Given S , sample the multiset of increments $\epsilon_1, \dots, \epsilon_n$ of i.i.d. $\mathcal{L}(\epsilon)$ variables conditional on their sum being S .
- 3 Given $\epsilon_1, \dots, \epsilon_n$, sample uniformly random permutation π , and form

$$W_k = S_k - (k/n)S \quad \text{where} \quad S_k = \sum_{j=1}^k \epsilon_{\pi(j)}.$$

Coupling Overview contd.: Induction

Use induction on n to couple $(W_k)_{1 \leq k \leq n}$ to a discrete Brownian bridge $\tilde{Z}_1, \dots, \tilde{Z}_n$ i.e. a mean zero Gaussian vector with covariances $(i \wedge j)(n - i \vee j)/n$:

- 1 Couple the marginals W_k and $\tilde{Z}_k \sim \mathcal{N}(0, k(n-k)/n)$ at time $k = \lfloor n/2 \rfloor$.
- 2 Use induction hypothesis to couple the left and right segment of the walk to the Brownian bridge.

Can check that the processes

$$S_i = W_i + (i/n)S \quad \text{and} \quad Z_i = \tilde{Z}_i + (i/n)Z$$

have the correct marginals and are coupled to have appropriate m.g.f. bound.

For X discrete, there doesn't exist T such that

$$E[Xf(X)] = E[Tf'(X)].$$

One can instead try to find an independent 'smoothing variable' Y so that there exist T such that

$$E[Xf(X + Y)] = E[Tf'(X + Y)].$$

Did not attain more: '...we do not know yet how to use Theorem 1.2 to prove the KMT theorem in its full generality, because we do not know how to generalize the smoothing technique of Example 3.'

Need smoothing in Step 1 to couple S_n and Brownian motion at time n , and in step 1 of the induction argument to couple W_k to a discrete Brownian bridge at time k .

Smoothing Lemma

For X Rademacher, $Y \sim \mathbb{U}(-1, 1)$ acts as a smoother satisfying

$$E[Xf(X + Y)] = E[(1 - XY)f'(X + Y)].$$

Can we get Y for general random variable X with mean zero and variance σ^2 ?

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Yes, indeed Y has the X -zero bias distribution i.e. Y satisfies

$$E[Xf(X)] = E[\sigma^2 f'(Y)]$$

Smoothing Lemma(Bhattacharjee, Goldstein)

If X is a mean zero random variable with finite, non-zero variance, and Y is an independent variable having the X -zero bias distribution, then

$$E[Xf(X + Y)] = E[(X^2 - XY)f'(X + Y)].$$

Generalizing to 'Smoothing Lemma' is perhaps not the largest obstacle.

'Hidden' Variance Parameter

For Rademacher, for $(B_t)_{0 \leq t \leq 1}$ a Brownian bridge, KMT follows from a bound on

$$E[\exp(\lambda \max_{1 \leq i \leq n} |W_i - \sqrt{n}B_{i/n}|)]$$

'Hidden' Variance Parameter

In general, KMT follows from a bound on

$$E[\exp(\lambda \max_{1 \leq i \leq n} |W_i - \gamma \sqrt{n} B_{i/n}|)]$$

where $\gamma^2 = \frac{1}{n} \sum_{i=1}^n \epsilon_i^2$.

For Rademacher, $\gamma^2 = 1$, as well as variance parameters γ_1^2, γ_2^2 for the left and right segments of the path.

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For Rademacher, $\gamma^2 = 1$, as well as variance parameters γ_1^2, γ_2^2 for the left and right segments of the path.

Problem: In the induction step of the coupling, the γ_1, γ_2 will not in general equal γ .

Effect of Variance Parameter

Recall in Rademacher case, bound on m.g.f. is given by

$$E \exp(\lambda \max_{0 \leq k \leq n} |W_k - \sqrt{n} B_{k/n}|) \leq \exp(C \log n) E \exp\left(\frac{K \lambda^2 S_n^2}{n}\right).$$

for $\lambda \leq \lambda_0$ where $S_n = \sum_{i=1}^n \epsilon_i$.

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In general becomes bound on m.g.f. of $\max_{0 \leq i \leq n} |W_i - \sqrt{n} \eta B_{i/n}|$, of the form

$$\exp(C \log n) E \exp\left(\frac{K_1 \lambda^2 S_n^2}{n} + K_2 \lambda^2 n (\gamma^2 - \eta^2)^2\right).$$

Recall $\sigma^2 > 0$ in Theorem 1.2 is arbitrary.

- 1 For Rademacher variables, S_n determines the multiset of increments $\epsilon_1, \dots, \epsilon_n$. In general, we need to sample from the conditional distribution of i.i.d. $\epsilon_1, \dots, \epsilon_n$ given they sum to S_n .
- 2 For X Rademacher, the 'smoothing' variable Y is known. Generalized Smoothing techniques necessary to use Theorem 1.2 for discrete variables.
- 3 Hidden 'variance parameter' $\gamma^2 = n^{-1} \sum_{i=1}^n \epsilon_i^2$ is identically 1 for Rademacher. In general, we need to have an additional term involving γ in the m.g.f. bound.

Existence of Marginal Coupling

One of the main tools in proving Theorem 1.2 is to find a 'nice' Stein matrix.

For W with Stein coefficient $h(W)$ and $Z \sim \mathcal{N}(0, \sigma^2)$, one can get a coupling of W and Z so that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^{2 \times 2}$ given by

$$T(x) = \begin{pmatrix} h(x_1) & \sigma \sqrt{h(x_1)} \\ \sigma \sqrt{h(x_1)} & \sigma^2 \end{pmatrix},$$

satisfies

$$E[X \cdot \nabla f(X)] = E[\text{Tr} \{ T(X) D^2 f(X) \}]$$

with $X_1 =_d W$ and $X_2 =_d Z$.

Multivariate Approach

Marginal argument appears ‘uncooperative’ and ‘brittle’.

Can avoid induction, relax conditions and handle dependence with multivariate version of Theorem 1.2 if given random vector $X \in \mathbb{R}^n$, one could produce ‘Stein matrix’ $T : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ p.s.d., uniformly bounded over $x \in \mathbb{R}^n$ such that

$$E[X \cdot \nabla f(X)] = E[\text{Tr} \{ T(X) D^2 f(X) \}].$$

Thank you